

Chapter 4 – Installing Ubuntu Server 22.04 LTS

Objective

In this chapter, we will install **Ubuntu Server 22.04.5 LTS** inside the VirtualBox virtual machine created in the previous chapter.

By the end of this chapter, you will have a working Ubuntu Server installation ready for post-installation configuration.

4.1 Starting the Installation

After attaching the Ubuntu Server ISO, start the virtual machine.

The Ubuntu boot menu will appear.

Select:

Try or Install Ubuntu Server

The installer will begin loading.

4.2 Select the Language

Choose your preferred installation language.

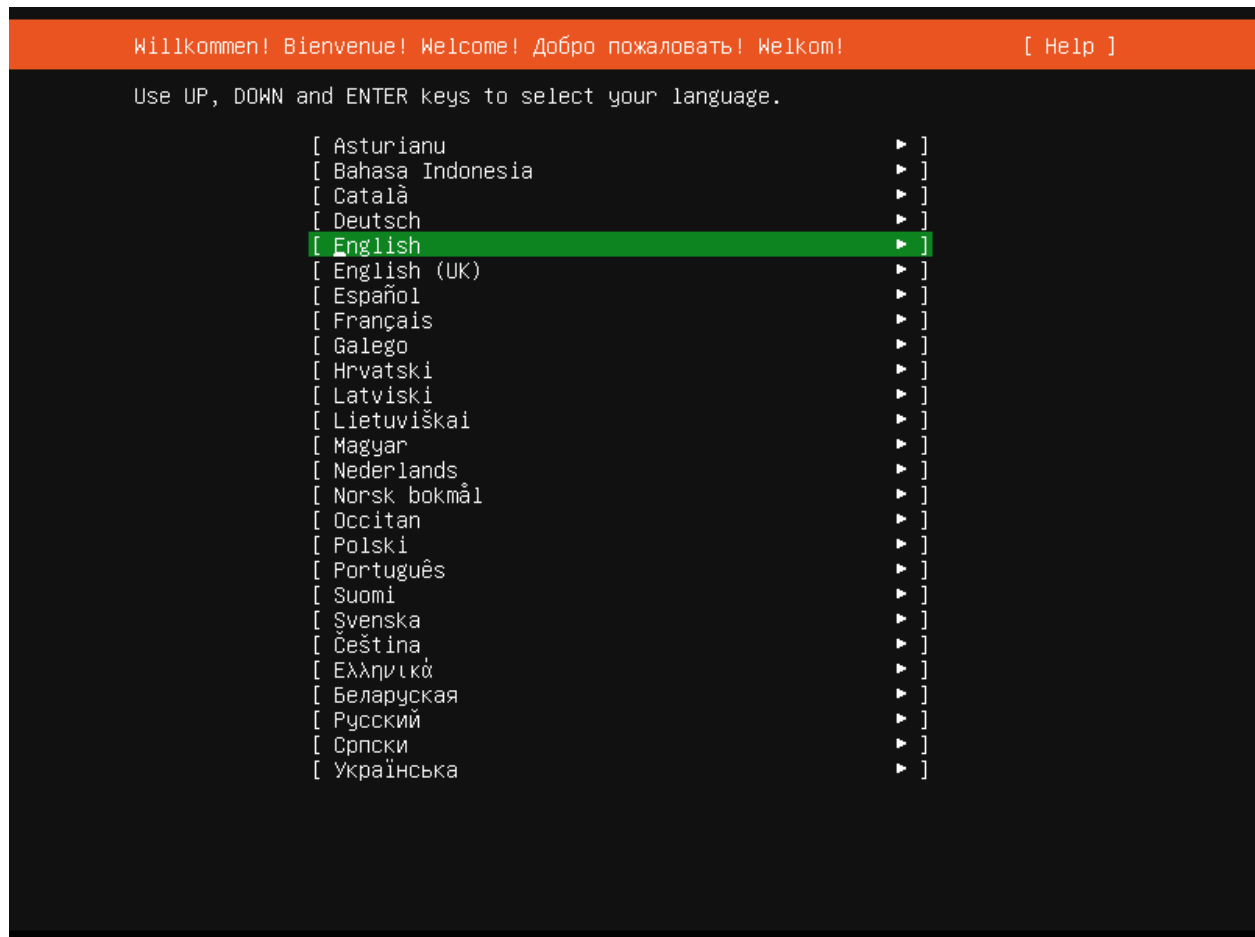
Example:

English

This language will be used during the installation process.

It can be changed later if required.

Screenshot: Ubuntu installer language selection.



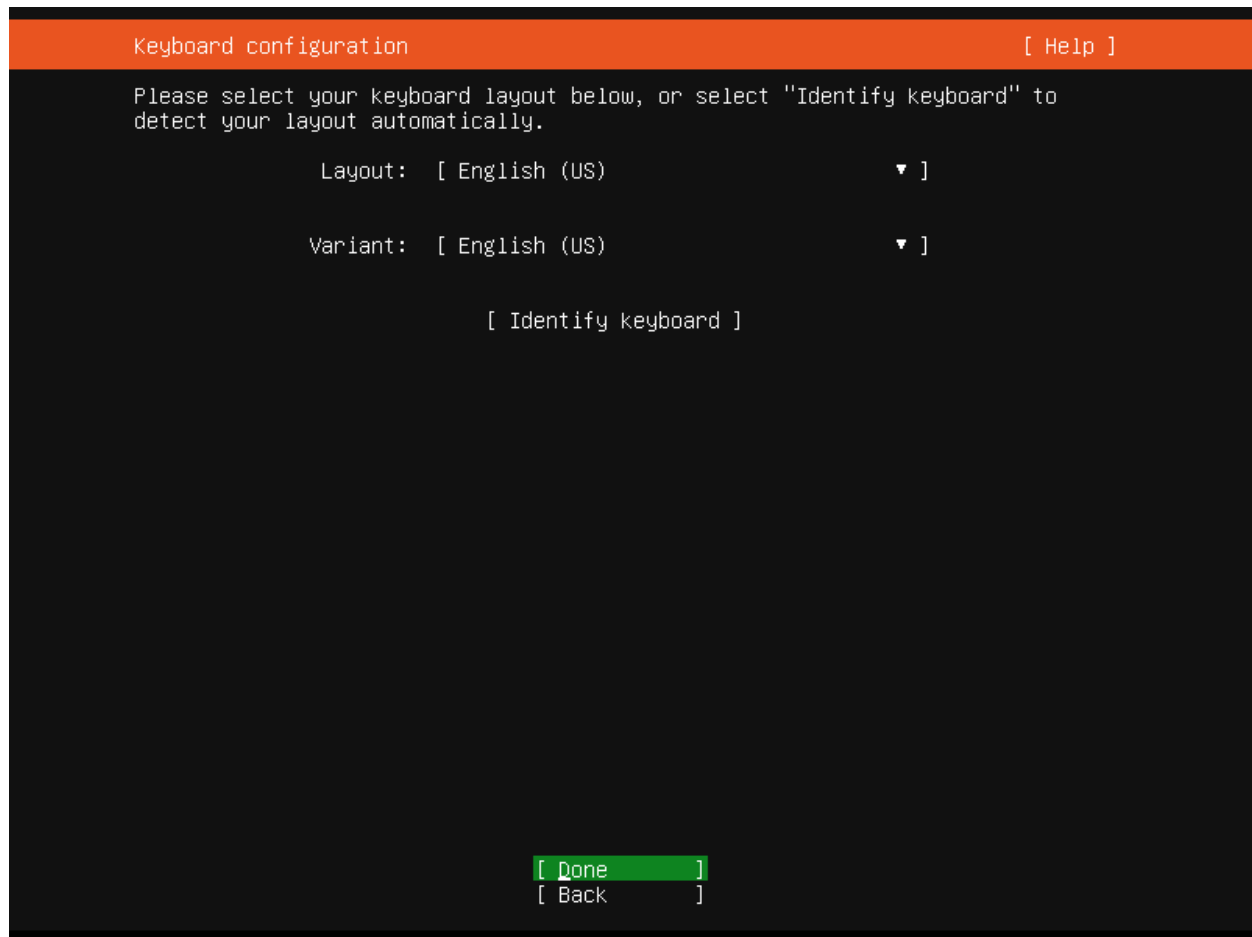
4.3 Select Keyboard Layout

Choose your keyboard layout.

Example:

English (US)

Selecting the correct layout ensures that special characters, symbols, and shortcuts work correctly.



4.4 Network Configuration

Ubuntu Server automatically detects available network adapters.

If your VM is connected using a **NAT adapter**, it should receive an IP address automatically through DHCP.

Example:

Interface:

enp0s3

IPv4:

10.0.2.15

At this stage, **accept the default DHCP configuration.**

A static IP address will be configured later.

4.5 Configure the Proxy (Optional)

Most home users do not use an HTTP proxy.

Leave this field empty.

Example:

Proxy:

<Leave Blank>

Select **Done**.

4.6 Ubuntu Archive Mirror

Ubuntu uses package mirrors to download software updates.

The installer automatically selects a default mirror.

Example:

<http://archive.ubuntu.com/ubuntu>

Unless you have a local mirror, accept the default and continue.

4.7 Storage Configuration

Select:

Use an entire disk

The installer will automatically create the required partitions.

For a home lab, the default layout is sufficient.

Guided storage configuration

[Help]

Configure a guided storage layout, or create a custom one:

☒ Use an entire disk

[VBOX_HARDDISK_VB5c13c56a-c70c7efa local disk 25.000G ▼]

☒ Set up this disk as an LVM group

☐ Encrypt the LVM group with LUKS

Passphrase:

Confirm passphrase:

☐ Also create a recovery key
The key will be stored as
~/recovery-key.txt in the live system and
will be copied to /var/log/installer/ in
the target system.

☐ Custom storage layout

[Done]

[Back]

4.8 Create Your User Account

Enter the required information.

Example:

Field	Example
Your Name	Kashif
Server Name	Ubunv22
Username	kashif
Password	*****

Choose a **strong password**, as this account will have administrative (sudo) privileges.

Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:
The name it uses when it talks to other computers.

Pick a username:

Choose a password:

Confirm your password:

[Done]

4.9 Install OpenSSH Server

When prompted, enable:

[X] Install OpenSSH Server

Why install OpenSSH?

OpenSSH allows secure remote administration of the server.

Instead of typing commands directly into the VirtualBox console, you can connect from another computer using SSH.

Example:

```
ssh kashifkhan@192.168.56.10
```

SSH is widely used by Linux administrators and cybersecurity professionals.

4.10 Featured Server Snaps

Ubuntu may offer optional packages such as:

- Docker
- Kubernetes
- PostgreSQL

For this guide:

Do not select any optional packages.

These can be installed later if needed.

4.11 Complete the Installation

The installer will:

- Copy system files
- Install Ubuntu Server
- Configure the bootloader
- Create the user account
- Finalize the installation

Installation time typically ranges from **5 to 15 minutes**, depending on your hardware.

Updating system [Help]

```
running 'curtin curthooks'
curtin command curthooks
  configuring apt
  configuring apt
  installing missing packages
  Installing packages on target system: ['grub-pc']
  configuring iscsi service
  configuring raid (mdadm) service
  configuring NVMe over TCP
  installing kernel
  setting up swap
  apply networking config
  writing etc/fstab
  configuring multipath
  updating packages on target system
  configuring pollinate user-agent on target
  updating initramfs configuration
  configuring target system bootloader
  installing grub to target devices
  copying metadata from /cdrom
final system configuration
calculating extra packages to install
installing openssh-server
  retrieving openssh-server
curtin command system-install
  unpacking openssh-server
curtin command system-install
  configuring cloud-init
  downloading and installing security updates
curtin command in-target |
```

[View full log]

4.12 Reboot the Server

When the installation finishes, select:

Reboot Now

If prompted to remove the installation media, simply press **Enter**.

The virtual machine will boot into the newly installed Ubuntu Server.

4.13 First Login

After rebooting, log in using the credentials created during installation.

Example:

Ubuntu 22.04.5 LTS

Username:

kashifkhan

Password:

After successful authentication, you should see the Linux shell prompt.

Example:

Kashif@ubunv22:~\$

This confirms that Ubuntu Server has been installed successfully.

```
[ 71.447454] cloud-init[1682]: Cloud-init v. 25.1.4-0ubuntu0~22.04.1 running 'modules:config' at Thu, 16 Jul 2026 05:31:35 +0000. Up 71.33 seconds.
[ 71.561008] cloud-init[1682]: Generating locales (this might take a while)...
[ 76.470924] cloud-init[1682]: en_US.UTF-8... done
[ 76.473075] cloud-init[1682]: Generation complete.
[ OK ] Finished Cloud-init: Config Stage.
Starting Cloud-init: Final Stage...
[ 77.601377] cloud-init[1713]: Cloud-init v. 25.1.4-0ubuntu0~22.04.1 running 'modules:final' at Thu, 16 Jul 2026 05:31:41 +0000. Up 77.48 seconds.
ci-info: no authorized SSH keys fingerprints found for user kashif.
<14>Jul 16 05:31:41 cloud-init: #####
<14>Jul 16 05:31:41 cloud-init: -----BEGIN SSH HOST KEY FINGERPRINTS-----
<14>Jul 16 05:31:42 cloud-init: 256 SHA256:ccy9SHGPFQFGx63U6gncAr3Nyx0ektnm/G104IemjM root@ubunv22 (ECDSA)
<14>Jul 16 05:31:42 cloud-init: 256 SHA256:HX7H6GvLIpr05KPQkFZzq03bpWKhhR0XJ0Ha6/qQKr8 root@ubunv22 (ED25519)
<14>Jul 16 05:31:42 cloud-init: 3072 SHA256:iqVOXPiknfK5As9ER4DYt1+8P5/wTjoq54bjse2cq7g root@ubunv22 (RSA)
<14>Jul 16 05:31:42 cloud-init: -----END SSH HOST KEY FINGERPRINTS-----
<14>Jul 16 05:31:42 cloud-init: #####
-----BEGIN SSH HOST KEY KEYS-----
ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBKIVQhVYVgJAcPwUQXpnqneuQ0jhN17qvq7EWvfGg5bTUIPkU1dTE1rgLoaIGNinQ0H0PkAy7sIObuKiPCMrZVc= root@ubunv22
ssh-ed25519 AAAAC3NzaC11ZDI1NTE5AAAAIK7LtkSMRTUkmRrPC5DqWEJ9248XyZfxgP6EFb+3UVH2+ root@ubunv22
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQgQC8CzS10I2b/KST+r/Ufb01v4Fkw0+H3hEc4I0d9UjWejXz12C02zsLdpQhGaxxsJjvxLHfgZDR3Vz3GM6vCCKVVAX0u5+gET6Ew/PY7YeB00w6Gx1ygz42efCGHJsbpwhCmJ0DiRBdw2Ef9gYH1u4ShFaJf8V/gB009PbXNxcgC1EsLpUDUwKdjEnX6eKy2GUBjE43sXM3r2wjXXrfgp67f7HEt6e/v1HFodi+p1p1B+uC/VknNKMuygUNKB8JyR2YkDULJNWpkkNyIMkPSDkzIiuZp9DaMao17RFEH7wPdwV30pIid1eZt26PSLCwdptYh1M4e2uY1Ck0+pQRF652NzqyyVwXQCWdm8Rh/QtHB9mpUYcXWnuqJ6UTZrzVLKkh6yRYKPBvoPK0SG+SkDfApURm/pz2UQ9FHRI88TS9oWqX+Qv8VzaMMUL1qBG9G0jBNWF i/qIgvDuEiD/oTilIakKnDzofkox2wBA0/5p7afYMrvXJMjZCRvVjNtPEck= root@ubunv22
-----END SSH HOST KEY KEYS-----
[ 77.916022] cloud-init[1713]: Cloud-init v. 25.1.4-0ubuntu0~22.04.1 finished at Thu, 16 Jul 2026 05:31:42 +0000. Datasource DataSourceNone. Up 77.90 seconds
[ OK ] Finished Cloud-init: Final Stage.
[ OK ] Reached target Cloud-init target.

kashif@ubunv22:~$
```

Best Practices

- Use a meaningful hostname (for example, wazuh-server).

- Enable OpenSSH during installation.
 - Use a strong password for the administrator account.
 - Keep the default partition layout unless you have advanced storage requirements.
 - Record your username and password securely.
-

Chapter Summary

In this chapter, you learned how to:

- Boot the Ubuntu Server installer.
 - Select the installation language and keyboard layout.
 - Configure networking using DHCP.
 - Configure storage.
 - Create an administrator account.
 - Install OpenSSH Server.
 - Complete the installation and perform the first login.
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One-Line Summary

Ubuntu Server 22.04.5 LTS is now installed and ready for post-installation configuration, including system updates, networking, and secure remote management.
